1.20 Inequalities: Using Graphs

Worked Solutions

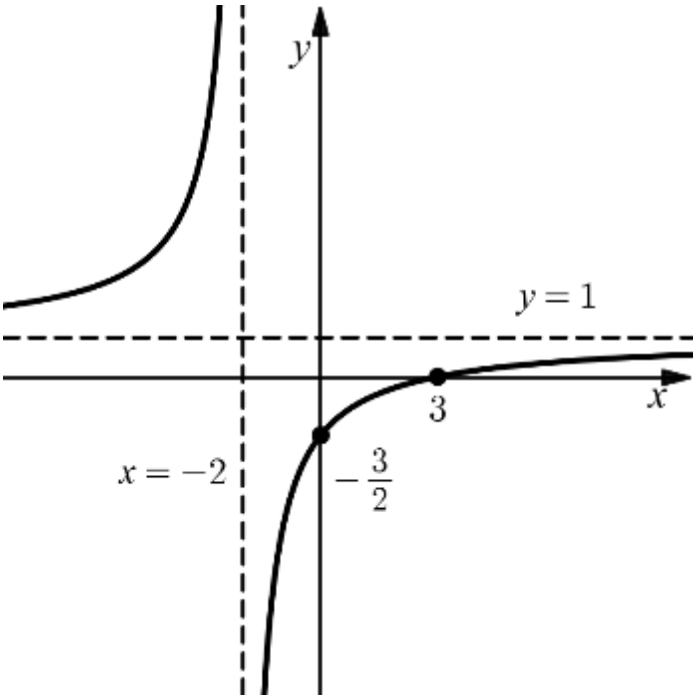
1 a)

$y = 1$ and $x = -2$

1 b)

When $x = 0$, $y = -\frac{3}{2}$

When $y = 0$, $x = 3$



1 c)

$\{x : x \in \mathbb{R}, -2 < x < 3\}$

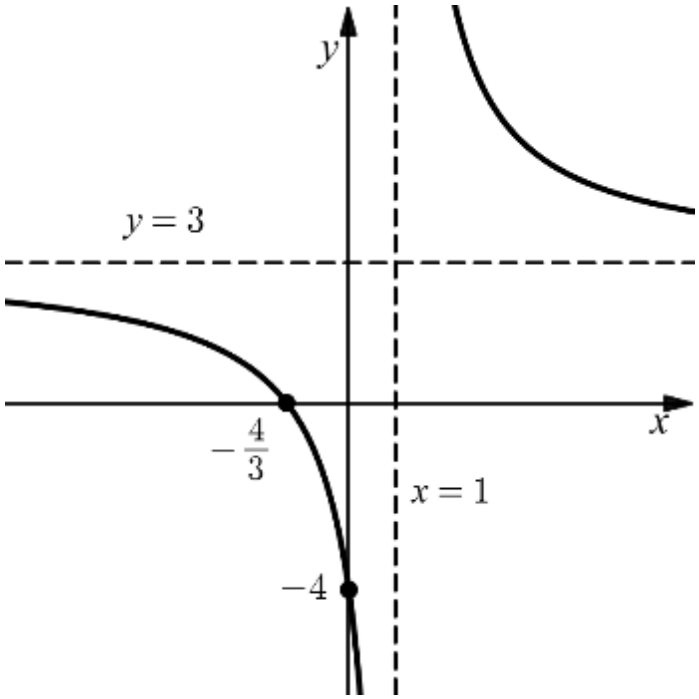
2 a)

$y = 3$ and $x = 1$

2 b)

When $x = 0$, $y = -4$

When $y = 0$, $x = -\frac{4}{3}$



2 c)

$x \in (\infty, -\frac{4}{3}] \cup (-1, \infty)$

3 a)

Asymptote of $x = 9$ gives $b = 9$

Using $(10, 15)$ then $15 = \frac{20-a}{10-9}$

$a = 5$ and $b = 9$

3 b)

When $x = 0$, $y = \frac{5}{9}$

When $y = 0$, $x = \frac{5}{2}$

$(\frac{5}{2}, 0)$ and $(0, \frac{5}{9})$

3 c)

$\frac{2x-5}{x-9} > 0$

$\{x : x \in \mathbb{R}, x < \frac{5}{2}\} \cup \{x : x \in \mathbb{R}, x > 9\}$

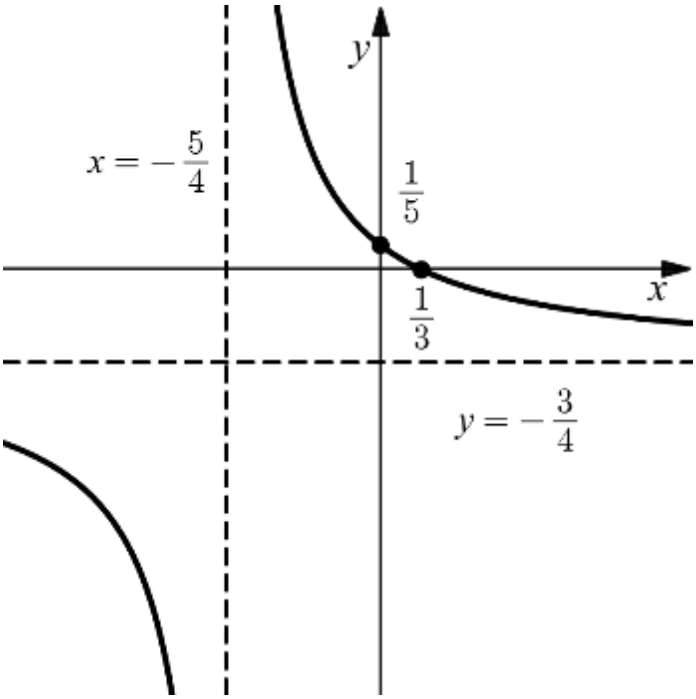
4 a)

$y = -\frac{3}{4}$ and $x = -\frac{5}{4}$

4 b)

When $x = 0$, $y = \frac{1}{5}$

When $y = 0$, $x = \frac{1}{3}$



4 c)

$$\frac{1-3x}{4x+5} = \frac{1}{2} \Rightarrow 2(1-3x) = 4x+5 \Rightarrow -3 = 10x \quad \therefore x = -\frac{3}{10}$$

$$\{x : x \in \mathbb{R}, x < -\tfrac{5}{4}\} \cup \{x : x \in \mathbb{R}, x > -\tfrac{3}{10}\}$$

5 a)

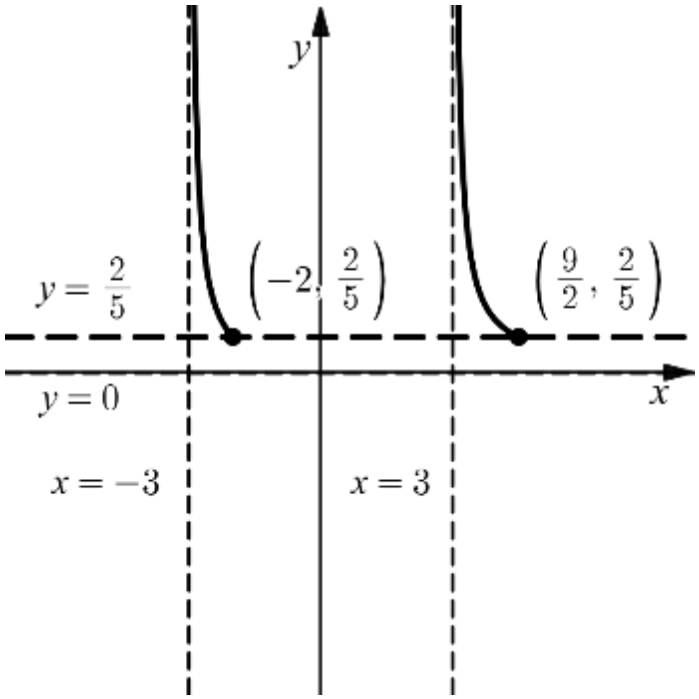
$k = 9$

5 b)

$$\frac{x}{x^2-9} = \frac{2}{5} \Rightarrow 5x = 2x^2 - 18 \Rightarrow 2x^2 - 5x - 18 = 0$$

$$\therefore x = -2 \text{ or } x = \frac{9}{2}$$

5 c)



$\{x : x \in \mathbb{R}, -3 < x \leq -2\} \cup \{x : x \in \mathbb{R}, 3 < x \leq \frac{9}{2}\}$

6 a)

$x^2 + x - 6 = (x + 3)(x - 2)$

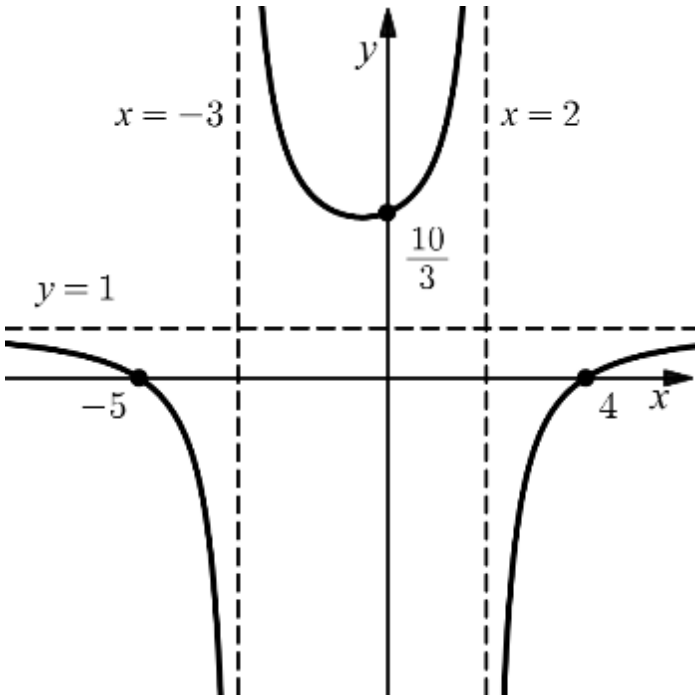
So asymptotes are

$$x = -3, \quad x = 2 \text{ and } y = 1$$

6 b)

When $x = 0$, $y = -\frac{10}{3}$

When $y = 0$, $x^2 + x - 20 = 0$ so $x = -5$ or $x = 4$



6 c)

$$\frac{x^2+x-20}{x^2+x-6}=\frac{9}{2}\Rightarrow 2x^2+2x-40=9x^2+9x-54\Rightarrow 7x^2+7x-14=0$$

so $x=1$ or $x=-2$

$$x\in(-\infty,3)\cup[-2,1]\cup(2,\infty)$$

7 a)

$a=2$

$$y=\frac{2x^2+4x+b}{x^2-3x+c}$$

At $(-1,0)$

$$0=2(-1)^2+4(-1)+b\quad\therefore b=2$$

$$y=\frac{2x^2+4x+2}{x^2-3x+c}$$

At $(3,\frac{16}{3})$

$$\frac{16}{3}=\frac{18+12+2}{9-9+c}\quad\therefore c=6$$

$a=2$, $b=2$ and $c=6$

7 b)

$$\frac{2x^2+4x+2}{x^2-3x+6}=2\Rightarrow 2x^2+4x+2=2x^2-6x+12\Rightarrow 10x=10\quad\therefore x=1$$

$$x\in(-\infty,1)$$

8 a)

$2x^2-4x+3=0$ has no real solutions

So the only asymptote is $y=\frac{1}{2}$

8 b)

$$\frac{x^2-1}{2x^2-4x+3}=k\Rightarrow x^2-1=k(2x^2-4x+3)$$
$$\Rightarrow(1-2k)x^2+4kx-(3k+1)=0$$

There exists solutions when:

$$(4k)^2+4(1-2k)(3k+1)\geq 0$$
$$16k^2-24k^2+4k+4\geq 0$$
$$2k^2-k-1\leq 0$$
$$(2k+1)(k-1)\leq 0 \text{ as req.}$$

8 c)

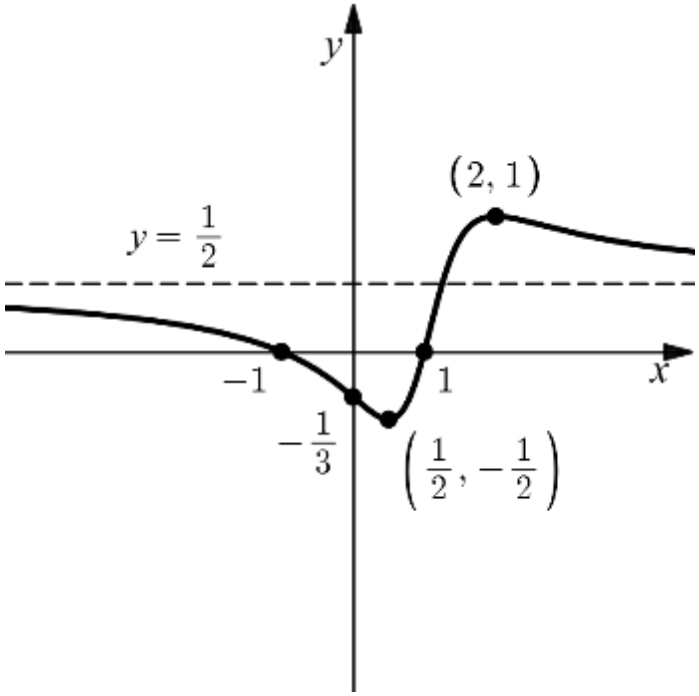
$$\text{When } y=-\frac{1}{2}, \quad -\frac{1}{2}=\frac{x^2-1}{2x^2-4x+3} \Rightarrow 2x^2-4x+3=2-2x^2$$
$$\Rightarrow 4x^2-4x+1=0 \text{ so } x=\frac{1}{2}$$
$$\text{When } y=1, \quad 1=\frac{x^2-1}{2x^2-4x+3} \Rightarrow 2x^2-4x+3=x^2-1$$
$$\Rightarrow x^2-4x+4=0 \text{ so } x=2$$

Min is at $(\frac{1}{2}, -\frac{1}{2})$, Max is at $(2, 1)$

8 d)

When $x=0$, $y=-\frac{1}{3}$

When $y=0$, $x=1$ or $x=-1$



8 e)

$$\frac{x^2-1}{2x^2-4x+3}=x-1 \Rightarrow x^2-1=(x-1)(2x^2-4x+3)$$
$$x^2-1=2x^3-4x^2+3x-2x^2+4x-3$$
$$2x^3-7x^2+7x-2=0$$

Let $\mathrm{f}(x)=2x^3-7x^2+7x-2$

$$\mathrm{f}(1)=2-7+7-2=0$$
$$\mathrm{f}(2)=16-28+14-2=0$$
$$\mathrm{f}(\tfrac{1}{2})=\tfrac{1}{4}-\tfrac{7}{4}+\tfrac{7}{2}-2=0$$

$$\therefore 2x^3 - 7x^2 + 7x - 2 = (x-1)(x-2)(2x-1) = 0$$

$$\therefore x = 1, \quad x = 2 \text{ or } x = \frac{1}{2}$$

$$\{x : x \in \mathbb{R}, x \leq \tfrac{1}{2}\} \cup \{x : x \in \mathbb{R}, 1 \leq x \leq 2\}$$

9 a)

Two asymptotes which are perpendicular so of the form $y = \frac{ax+b}{cx+d}$

The asymptotes give: $y = \frac{-2x+k}{x-1}$

At $(3, 0)$ then $0 = \frac{-6+k}{3-1}$ so $k = 6$

$$\mathrm{f}(x) = \frac{-2x+6}{x-1}$$

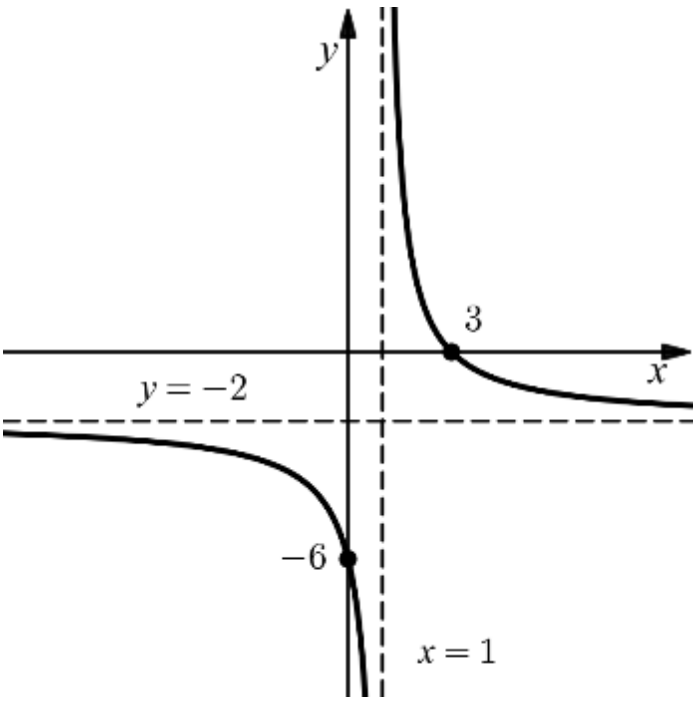
9 b)

When $x = 0$, $y = -6$

When $y = 0$, $x = 3$

$$(0, -6) \text{ and } (3, 0)$$

9 c)



9 d)

$$\frac{-2x+6}{x-1} = 4 \Rightarrow -2x + 6 = 4x - 4 \Rightarrow 6x = 10$$

$$\therefore x = \frac{5}{3}$$

$$\therefore x \in (1, \tfrac{5}{3}]$$

10 a)

The curve is of the form $y = \frac{3x^2+ax+b}{x^2+cx+2}$

Consider the points:

$$-1 = \frac{b}{2} \text{ so } b = -2$$

But also

$$\frac{29}{7} = \frac{3a+25}{3c+11} \Rightarrow 319 + 87c = 21a + 175 \Rightarrow 21a - 87c = 144$$

$$-5 = \frac{1-a}{3-c} \Rightarrow -15 + 5c = 1 - a \Rightarrow a + 5c = 16$$

$a = 11$, $c = 1$

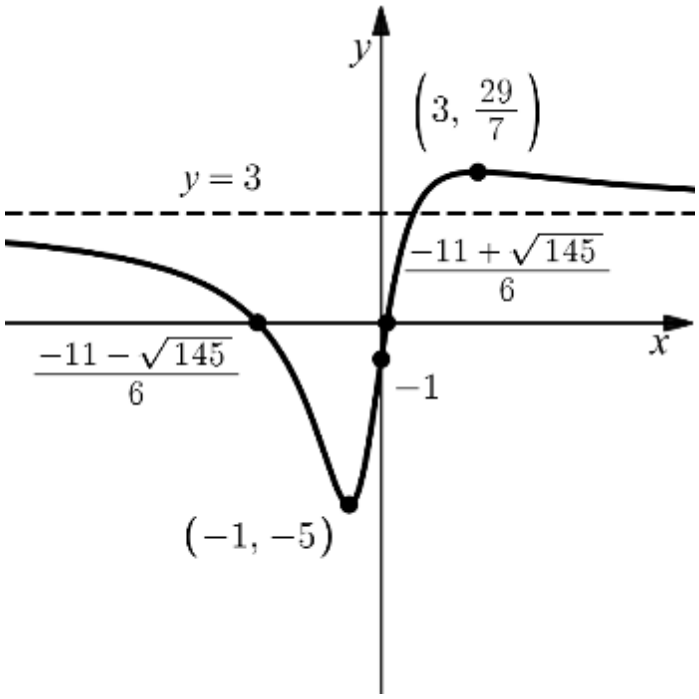
$$\therefore \mathrm{f}(x) = \frac{3x^2 + 11x - 2}{x^2 + x + 2}$$

10 b)

When $y = 0$, $3x^2 + 11x - 2 = 0$

$$\therefore x = -\frac{11}{6} + \frac{\sqrt{145}}{6} \text{ or } x = -\frac{11}{6} - \frac{\sqrt{145}}{6}$$

10 c)



10 d)

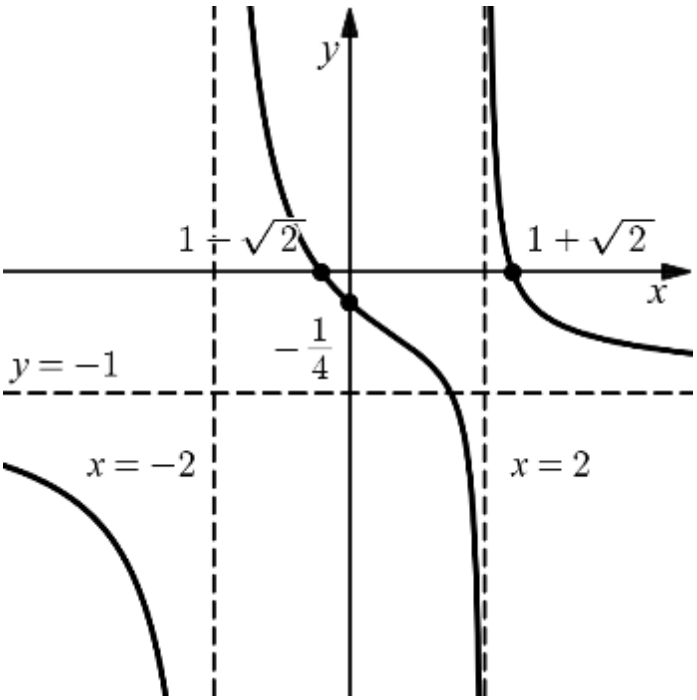
$$-1 = \frac{3x^2 + 11x - 2}{x^2 + x + 2} \Rightarrow -x^2 - x - 2 = 3x^2 + 11x - 2$$

$$\Rightarrow 4x^2 + 12x = 0 \Rightarrow 4x(x+3) = 0$$

$\therefore x = 0$ or $x = -3$

$$\{x : x \in \mathbb{R}, x \leq -3\} \cup \{x : x \in \mathbb{R}, x \geq 0\}$$

11 a)



11 b)

Assume the function is of the form: $y = \frac{ax^2+bx+c}{px^2+qx+r}$

$(0, -\frac{1}{4})$ gives that $\frac{c}{r} = -\frac{1}{4}$

Numerator: $a(x-(1+\sqrt{2}))(x-(1-\sqrt{2})) = a(x^2-2x-1)$

Denominator: $p(x-2)(x+2) = p(x^2-4)$

$y = -1$ gives that $\frac{a}{p} = -1$

$$\therefore a = -p$$

$$\frac{a(x^2-2x-1)}{-a(x^2-4)} = -\frac{x^2-2x-1}{x^2-4}$$

$$\therefore \mathrm{f}(x) = -\frac{x^2-2x-1}{x^2-4}$$

11 c)

$$-1 = -\frac{x^2-2x-1}{x^2-4} \Rightarrow x^2-4 = x^2-2x-1 \Rightarrow 2x = 3$$

$$\therefore x = \frac{3}{2}$$

$$\therefore x \in (-\infty, -2) \cup [\tfrac{3}{2}, 2)$$